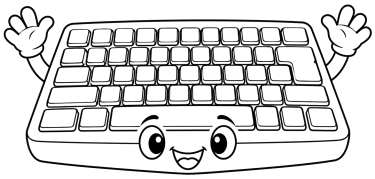
TEACHER GUIDE

ANSWERS AND MARKS

GRADE 9 · INFORMATION AND COMMUNICATION TECHNOLOGY

UNIT 2
Electronic Spreadsheets



! FOR THE TEACHER ONLY

This booklet has the answers to every exercise in the Unit 2 Student Workbook, in the same order. The exercise letters match the workbook.

i WHAT IS IN THIS GUIDE

- **Marks and pacing** - every exercise and its marks
- **Class marks record** - one page to copy for each class
- **Practicals at a glance** - Practicals 1 - 10, a quiz, a practical test
- **Answers** - chapters 1 to 5, in the same order as the workbook
- **Unit 2 Test** - the marking scheme and the common mistakes
- **Extra** - textbook links and notes, homework ideas, teaching tips

★ BEFORE THE FIRST LESSON

- ☐ MS Excel or LibreOffice Calc opens on every computer.
- ☐ The projector shows a worksheet clearly from the back of the room.
- ☐ A folder is ready for each class to save the work.

"Students who type the formula themselves never forget it."

- THE KING OF ICT

TEACHER GUIDE

Marks and Pacing

You decide how much class time each part needs

Chapter	Exercise	Marks
Chapter 1	A Fill in the blanks	8
	B Name the tool	10
	C True or false	6
	D Match them	6
	E Which tool would you use?	6
	F Choose the correct answer	6
Chapter 2	A Label the worksheet	6
	B Ranges in a worksheet	6
	C Fill in the blanks	8
	D Read the worksheet	6
	E Write the cell addresses	8
	F True or false	6
	G Choose the correct answer	4

Chapter	Exercise	Marks
Chapter 3	A Which tool was used?	8
	B What will the cell show?	6
	C True or false	6
	D Name the tools	8
	E Which number format?	7
	F Which tool would you use?	8
	G Choose the correct answer	4
Chapter 4	A Match the symbols	5
	B Write the formulas	8
	C Find the results	8
	D Results from a worksheet	8
	E Which function?	6
	F Work them out	8
	G Spot the mistakes	4
Chapter 5	A Name the charts	6
	B Read the chart	6
	C Choose the correct answer	4
	D Crossword	10
End	Unit 2 Test	40
TOTAL		**246**

! SUGGESTED SHAPE OF ONE CHAPTER

READ	Teach the chapter pages. Students read with you and underline the key words.
DO	The practical tasks on real computers, in pairs.
WRITE	The written exercises. Walk around and help.
MARK	Mark the work together. Fill the tick boxes.

★ TEACHING TIPS

- Show the textbook pictures (pages 23 - 27) on the projector and compare them with the drawings in the workbook.
- Let pairs share a computer: one types, one watches the Name box and the Formula Bar. Swap every ten minutes.

TEACHER GUIDE

Class Marks Record

Copy this page for each class

No	Student name	Ch 1 (42)	Ch 2 (44)	Ch 3 (47)	Ch 4 (47)	Ch 5 (26)	Test (40)	Total (246)
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								
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16								
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22								
23								
24								
25								
26								
27								
28								
29								

TEACHER GUIDE

Practicals at a Glance

Workbook activities 2.1 - 2.10 (textbook page 28)

Practical	Students do	Check that ...
1 (2.1)	start the program; workbook and worksheets	they find the sheet tabs and add a worksheet.
2 (2.2)	New, Save, Open, Print Preview, Print	the file My Friends opens again after it is closed.
3 (2.3)	Spelling, Copy, Cut, Paste, Undo, Redo	they can say the difference between Cut and Copy.
4 (2.4)	the Name box, the Formula Bar, selecting	the Name box shows G7; they select B, row 3 and D5:E10.
5 (2.5)	font, size, bold, italic, underline, colours, borders	A1 is Cambria 16, bold, with a fill colour; A1:B5 has all borders.
6 (2.6)	alignment, wrap text, orientation, merge	Sports Meet is merged and centred over A1:D1.
7 (2.7)	number formats and decimals	0.25 shows 25% and 5677 shows 5.68E+03.
8 (2.8)	formulas with = + - * /	C1:C4 show 30, 20, 125, 5; with A1 = 40 they show 45, 35, 200, 8.
9 (2.9)	SUM, AVERAGE, MAX, MIN	B7:B10 show 16000, 3200, 4100, 2500; with March = 3300: 16500, 3300, 4100, 2500.
10 (2.10)	column, line, pie, bar, area and scatter charts	each chart shows the five months.

! SAFETY AND MATERIALS

- Computers with MS Excel or LibreOffice Calc. Test the program on every computer before the lesson.
- A projector for the demonstrations. Prepare the trip fund and the stationery shop worksheets as files.
- A folder for each class to save the work. No food or drinks near the computers.

Textbook links

Workbook	Textbook pages	Activities
Chapter 1 - Spreadsheets and their common tools	22 - 23	2.1 - 2.3
Chapter 2 - Inside a worksheet	24	2.4
Chapter 3 - Formatting the cells	25 - 26	2.5 - 2.7
Chapter 4 - Formulas and functions	27	2.8, 2.9
Chapter 5 - Charts	27 - 28	2.10
Unit 2 in One Page	28 (summary)	-

★ TEACHING TIP

- Say "column letter first, then row number" every time you write a cell address on the board.

! NOTES ON THE TEXTBOOK

- Page 24: "Intersection between a raw and a column" - read **row**. "The row letter and the column number ... cell Address" - it is the **column letter** and then the **row number** (G7).
- Page 25: "To apply boarders" - read **borders**.
- Page 26: 25 February 2018 was a **Sunday**, not a Tuesday. The workbook shows **Sunday, February 25, 2018**, as a spreadsheet does.
- Page 27: "display spreadsheet date graphically" - read **data**. Page 28: "raw number 1" - read **row**.
- Page 28 sends students to Activities 2.1 - 2.10 of the workbook. Here they are Practicals 1 - 10, one for each part of the chapter (the numbers on each DO IT box).

? QUICK ORAL QUIZ - START OR END A LESSON WITH IT

Ask the class ...	Answer
1. What is a spreadsheet file called?	**a workbook**
2. Which tool cancels a performed action?	**Undo**
3. What does the Name box show?	**the selected cell**
4. The cell in column D and row 5?	**D5**
5. How many cells are in B2:C4?	**6**
6. Which tool makes cells into one larger cell?	**Merge and Center**
7. 0.25 with the Percentage format?	**25%**
8. What does every formula begin with?	**=**
9. Which function finds the highest value?	**MAX**
10. Which chart is a circle cut into slices?	**Pie chart**

> PRACTICAL TEST - 30 MINUTES, 20 MARKS (OPTIONAL)

Totals: 195, 221, 166, 265, 184. Averages: 65.0, 73.7, 55.3, 88.3, 61.3. ICT: highest 92, lowest 45.

Step	The student must ...	Marks
1	Type the marks sheet of the exercise "Find the results" (A1:D6).	3
2	Totals in E2:E6 with **SUM** (one formula for each student).	3
3	Averages in F2:F6 with **AVERAGE**, shown with 1 decimal place.	3
4	The highest and the lowest ICT marks with **MAX** and **MIN**.	2
5	Row 1: Bold, a Fill colour and Borders. A title merged and centred above the table.	3
6	Any one number format or alignment tool used on the marks.	2
7	A column chart of the names and the totals.	3
8	Save the file with your name.	1
	Total	**20**

ANSWERS

Chapter 1

Spreadsheets and their common tools

A Fill in the blanks

8 marks

1. accounting	**2.** accountants	**3.** data	**4.** tabular
5. cells	**6.** charting	**7.** workbook	**8.** worksheet

B Name the tool

10 marks

1. Save	**2.** Undo	**3.** Print	**4.** Copy	**5.** New
6. Spelling	**7.** Cut	**8.** Print Preview	**9.** Open	**10.** Redo

1 mark each. Paste is the extra word in the box.

C True or false

6 marks

1. **T - A spreadsheet is an electronic version of paper based accounting sheets.**
2. **T - An electronic spreadsheet stores data in a tabular form.**
3. **F - A workbook can have only one worksheet.**
4. **F - Print Preview prints the document on paper.**
5. **T - Redo does an undone action again.**
6. **F - Copy removes an item from its current location.**

3: a workbook may contain multiple worksheets. 4: Print Preview only shows how the document will look; Print prints it. 6: Cut removes an item; Copy keeps it.

D Match them

6 marks

1. New → C	**2.** Open → E	**3.** Print Preview → F
4. Undo → D	**5.** Cut → A	**6.** Spelling → B

E Which tool would you use?

6 marks

1. See how your worksheet will look on paper before you print it: **Print Preview**
2. Keep your workbook in the computer to use it again later: **Save**
3. Work again on a workbook that you saved yesterday: **Open**
4. Cancel the last thing you did: **Undo**
5. Move a word from one cell to another cell: **Cut, then Paste**
6. Check the spelling of the words in your worksheet: **Spelling**

F Choose the correct answer

6 marks

1. (b) New	**2.** (b) multiple worksheets
3. (c) Undo	**4.** (a) cells
5. (b) Paste	**6.** (d) cooking

ANSWERS

Chapter 2

Inside a worksheet

A Label the worksheet

6 marks

1. Name box	**2.** Formula Bar	**3.** Column heading
4. Row heading	**5.** Cell (G7, the selected cell)	**6.** Cell range (D5:E10)

1 mark each. Sheet tab and Title Bar are the extra words.

B Ranges in a worksheet

6 marks

1. B2:E2	**2.** G2:G6	**3.** B4:D6
4. 4	**5.** 9	**6.** R2

1 mark each. R1 is 4 columns of row 2. R3 is 3 columns x 3 rows = 9 cells.

C Fill in the blanks

8 marks

1. cell	**2.** typed	**3.** A	**4.** 1
5. column	**6.** range	**7.** heading	**8.** number

D Read the worksheet

6 marks

1. What is in cell A4? **Kasun**

2. What is in cell C5? **90**

3. Write the address of the cell with Gemunu. **B3**

4. Write the cell range of all the marks. **C2:C6**

5. You click on the mark of Ravi. What does the Name box show? **C6**

6. You click on the mark of Tharushi and type 95. What does the Formula Bar show? **95**

TIP: Play "Find the cell": call out an address such as D7 and let the class point to it on the projector.

E Write the cell addresses

8 marks

1. star: **B2**	**2.** black circle: **E1**	**3.** square: **G4**	**4.** triangle: **A5**
5. diamond: **D3**	**6.** heart: **F6**	**7.** ring: **C6**	**8.** cross: **E5**

1 mark each: the column letter first, then the row number.

F True or false

6 marks

1. **T - The cell address G7 means column G and row 7.**
2. **F - In a cell address, the row number comes first.**
3. **F - The Name box shows what is being typed in a cell.**
4. **T - To select a whole row, click its row heading.**
5. **T - The cell range B2:B5 has 4 cells.**
6. **F - The cell range A1:C2 has 5 cells.**

2: the column letter comes first (G7). 3: the Name box shows the selected cell; the Formula Bar shows what is typed. 6: A1:C2 has 3 x 2 = 6 cells.

G Choose the correct answer

4 marks

1. (b) D4	**2.** (d) 6
3. (b) the selected cell	**4.** (a) the row heading

TEACHING TIPS

- Test MS Excel or LibreOffice Calc on every computer before the first practical.
- Show that Cut moves an item and Copy keeps the original. Ask: where is the text now?
- For a cell range, count the columns and the rows and multiply: D5:E10 is 2 x 6 = 12 cells.

ANSWERS

Chapter 3

Formatting the cells

A Which tool was used?

8 marks

1. Right align	**2.** Centre align	**3.** Top	**4.** Bottom
5. Middle	**6.** Wrap Text	**7.** Orientation	**8.** Merge and Center

1 mark each. Centre align moves the content between left and right; Middle moves it between top and bottom.

B What will the cell show?

6 marks

1. 0.75, Percentage → **75%**	**2.** 12500, Scientific → **1.25E+04**
3. 250, Currency → **Rs. 250.00**	**4.** 0.5, Fraction → **1/2**
5. 3/10/2025, Long Date → **Monday, March 10, 2025**	**6.** 7.25, Decrease decimals → **7.3**

1 mark each. 7.25 with one decimal place rounds up to 7.3.

C True or false

6 marks

1. **T - Italic makes the letters slanted.**
2. **F - Fill colour changes the colour of the letters.**
3. **T - Wrap Text makes all the content visible within a cell.**
4. **T - Merge and Center combines adjacent cells into one larger cell.**
5. **F - Orientation changes the size of the letters.**
6. **F - The Short Date format shows a date as DD/MM/YYYY.**

2: Fill colour changes the background; Color changes the letters. 5: Orientation changes the direction. 6: the Short Date is MM/DD/YYYY (textbook page 26).

D Name the tools

8 marks

1. Bold	**2.** Fill colour	**3.** Italic	**4.** Borders
5. Underline	**6.** Color (font colour)	**7.** Merge and Center	**8.** Wrap Text

1 mark each. Orientation and Size are the extra words.

E Which number format?

7 marks

1. Rs. 75.00 → **Currency**	**2.** 35% → **Percentage**
3. 3/4 → **Fraction**	**4.** 4.50E+06 → **Scientific**
5. 08/15/2025 → **Short Date**	**6.** Friday, August 15, 2025 → **Long Date**
7. 10:00 a.m. → **Time**	

★ TEACHING TIPS

- Use a school example for formatting: the sports meet results with the house names in colour.
- Fill colour colours the cell; Color colours the letters. Let students try both on the same cell.
- Centre align works from left to right; Middle works from top to bottom. Make row 3 tall to show it.

F Which tool would you use?

8 marks

1. Make the heading darker than normal: **Bold**
2. Change the background colour of a cell: **Fill colour**
3. Draw lines around the cells of a table: **Borders**
4. Make the letters slanted: **Italic**
5. Join A1:E1 into one cell with the heading in the centre: **Merge and Center**
6. Show a long sentence completely inside one cell: **Wrap Text**
7. Change the colour of the letters: **Color (font colour)**
8. Change the direction of the text: **Orientation**

G Choose the correct answer

4 marks

1. (b) Bold	**2.** (b) Fill colour
3. (b) 45%	**4.** (c) Long Date

ANSWERS

Chapter 4

Formulas and functions

A Match the symbols

5 marks

1. = → **B**	**2.** + → **D**	**3.** - → **E**	**4.** * → **C**	**5.** / → **A**

B Write the formulas

8 marks

1. D2: **=B2*C2** gives **180**	**2.** D4: **=B4*C4** gives **280**
3. D6: **=D2+D3+D4+D5** gives **690**	**4.** B7: **=B4-B2** gives **10**

1 mark for the formula, 1 mark for the result. In D6 also accept =SUM(D2:D5); in D2 accept =C2*B2.

C Find the results

8 marks

1. =SUM(B2:B6) = **352**	**2.** =AVERAGE(B2:B6) = **70.4**
3. =MAX(C2:C6) = **85**	**4.** =MIN(D2:D6) = **58**
5. =SUM(B2:D2) = **195**	**6.** =AVERAGE(B2:D2) = **65**
7. =MAX(B2:D6) = **92**	**8.** =MIN(B2:D6) = **45**

1 mark each. AVERAGE(B2:B6) = 352 / 5 = 70.4.

D Results from a worksheet

8 marks

1. =A1+B1 = **16**	**2.** =A2-C2 = **12**
3. =B2*C1 = **15**	**4.** =A2/B2 = **4**
5. =A1+B1+C1 = **19**	**6.** =B3*C3 = **20**
7. =(A3+B3)/C3 = **8**	**8.** =(A2-B3)*C1 = **30**

1 mark each. =(A3+B3)/C3 is (6 + 10) / 2.

E Which function?

6 marks

You want to find ...	SUM	AVERAGE	MAX	MIN
The total money collected for the school trip	☑	☐	☐	☐
The highest mark in the ICT test	☐	☐	☑	☐
The average rainfall of a month	☐	☑	☐	☐
The lowest price in a price list	☐	☐	☐	☑
The total runs of a cricket team	☑	☐	☐	☐
The average age of the students of a class	☐	☑	☐	☐

F Work them out

8 marks

1. =15+27 = **42**	**2.** =50-18 = **32**
3. =6*8 = **48**	**4.** =72/8 = **9**
5. =(4+6)*5 = **50**	**6.** =(20-8)/4 = **3**
7. =2*(15+5) = **40**	**8.** =(9+11)/(7-2) = **4**

1 mark each. The brackets are worked out first.

G Spot the mistakes

4 marks

1. plus (+) → **equals (=)**	**2.** x → *****
3. total → **average**	**4.** lowest → **highest**

★ TEACHING TIPS

- A number format changes only how a value looks. Type =A2*2 next to a percentage cell to prove it.
- Read the textbook date with the class: 02/25/2018 is month, day, year (MM/DD/YYYY).
- Ask students to predict the result of a formula before they press Enter. Then check it together.
- Save often (Ctrl + S). A power cut can take away a whole lesson of work.

ANSWERS

Chapter 5

Charts

A Name the charts

6 marks

1. Pie chart	**2.** Bar chart	**3.** Scatter chart
4. Column chart	**5.** Area chart	**6.** Line chart

B Read the chart

6 marks

1. What type of chart is this? **Column chart**

2. Which house collected the most bottles? **Gemunu**

3. Which house collected the fewest bottles? **Tissa**

4. How many bottles did Vijaya collect? **40**

5. How many more bottles did Gemunu collect than Tissa? **30**

6. Which function finds the total number of bottles? What is the total? **SUM; 180**

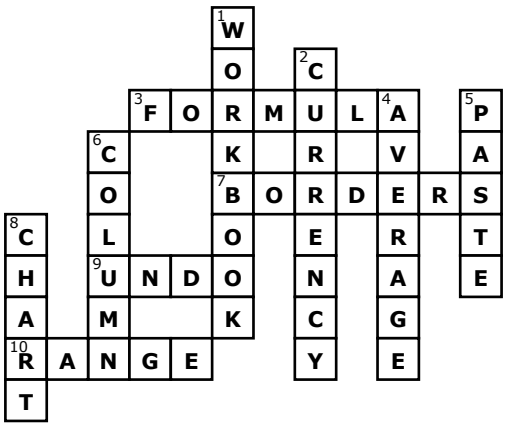
C Choose the correct answer

4 marks

1. (b) graphically	**2.** (c) Pie chart
3. (b) lie from left to right	**4.** (d) Borders

D Crossword

10 marks



Across: 3 FORMULA, 7 BORDERS, 9 UNDO, 10 RANGE

Down: 1 WORKBOOK, 2 CURRENCY, 4 AVERAGE, 5 PASTE, 6 COLUMN, 8 CHART

TEACHING TIP

- Change one value in a worksheet with formulas and let the class watch every result change.

MARKING SCHEME

Unit 2 Test

Total 40 marks

1 Choose the correct answer

8 marks (1 each)

1. (a) paper based accounting sheets	**2.** (b) Undo
3. (b) the currently selected cell	**4.** (c) 8
5. (a) Wrap Text	**6.** (c) Scientific
7. (b) 30	**8.** (a) A, B only

2 Fill in the blanks

8 marks (1 each)

1. worksheets	**2.** A	**3.** cell	**4.** Formula Bar
5. Fill colour	**6.** =	**7.** 9	**8.** /

3 Name the tools

8 marks (1 each)

1. Save	**2.** Paste	**3.** Borders	**4.** Wrap Text
5. Merge and Center	**6.** Percentage	**7.** Pie chart	**8.** Column chart

Accept small spelling mistakes if the tool is clear (for example "Merge and Centre").

4 Answer these

16 marks

1. Write a formula for D2 to add the track and field points of Vijaya. What does D2 show? (2 marks)

=B2+C2 (1 mark). It shows 149 (1 mark).

2. Write a function for B6 to find the total track points of all the houses. What does it show? (2 marks)

=SUM(B2:B5) (1 mark). It shows 313 (1 mark).

3. Write a function for B7 to find the average track points. What does it show? (2 marks)

=AVERAGE(B2:B5) (1 mark). It shows 78.25 (1 mark).

4. Write a function to find the highest field points. What does it show? (2 marks)

=MAX(C2:C5) (1 mark). It shows 81 (1 mark).

5. Write the cell range of the field points. How many cells are in it? (2 marks)

C2:C5 (1 mark); 4 cells (1 mark).

6. You click on cell A3. What does the Name box show? What is in A3? (2 marks)

A3 (1 mark); Gemunu (1 mark).

7. Name two tools you would use to make row 1 stand out as a heading row. (2 marks)

Any two: Bold, Size, Fill colour, Color (font colour), Borders, Underline, Italic, Centre align. 1 mark each.

8. Which chart would you use to compare the total points of the houses? Write its first step. (2 marks)

A column chart (or a bar chart) (1 mark). First select the data (the house names and the totals), then Insert - Column (1 mark).

WHAT STUDENTS USUALLY GET WRONG

- They write the row number first (7G). A cell address is the column letter, then the row number: **G7**.
- They mix up the Name box (the selected cell) and the Formula Bar (what is typed in the cell).
- They forget the = sign, or type x for multiply. The symbols are **+ - * /**.
- They count the cells of a range wrongly: B2:C5 is 2 columns x 4 rows = **8** cells.
- They mix up Fill colour (the background) and Color (the letters), and Centre align and Middle.
- They think a number format changes the value. 0.25 shown as 25% is still 0.25 in the cell.
- They select only the numbers for a chart. Select the labels too, so that the chart shows the names.

HOMEWORK AND EXTRA PRACTICE

- Make a worksheet of a family's spending for one week (rice, bus fare, electricity ...) with SUM, AVERAGE, MAX, MIN and a column chart.
- Collect the marks of five friends in three subjects. Find each total with SUM and draw a chart.
- Draw a worksheet from memory and label the Name box, the Formula Bar, a column, a row, a cell and a cell range.
- Fast learners: format a class time table with Merge and Center, Wrap Text, Borders and Fill colour.

TEACHING TIP

- Use real class data: trip savings, test marks, house points. Students love to see their own numbers.